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Teaching AI agents at university

A 14-week, 3-credit module in intelligent agents for third- or fourth-year computer science students. Prerequisites: algorithms, probability and programming. Darsi lecture 1 anchors weeks 1 and 5, lecture 2 weeks 3 and 8, lecture 3 week 9 and lecture 4 weeks 12 and 13.

Learning outcomes

1. Specify agents and environments with PEAS and MDPs.
2. Compute optimal policies for small MDPs and design rewards that resist gaming.
3. Implement heuristic search and tool-using language-model agents.
4. Evaluate agents rigorously with held-out tasks, repeated runs and intervals.
5. Design agents with least privilege, human approval and accountability.

14-week syllabus

Week	Topic	In class and out
1	Agents, environments and rationality	PEAS exercise
2	Uninformed and heuristic search	Problem set 1
3	Classical planning	Lab 1: A* and a small planner
4	Probability and decisions	Problem set 2
5	Markov decision processes	Lab 2: reward design
6	Reinforcement learning	Problem set 3
7	Language models as agents	Midterm exam
8	Tool use and ReAct	Lab 3: a tool-using agent
9	Evaluating agents	Evaluation protocol
10	Multi-agent systems	Problem set 4
11	Memory and retrieval	Project proposal and risk review
12	Safety, permissions and oversight	Lab 4: approval gates
13	Governance and accountability	Project checkpoint
14	Project presentations and review	Final exam

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Lecture slide outline

Lecture 1: Agents and MDPs

- A cleaning robot that made a mess
- PEAS and environment types
- States, actions, rewards, γ
- Specification gaming
- Activity: break the reward

Lecture 2: Planning and tools

- An assistant that invented a room
- STRIPS and A*
- Admissible heuristics
- ReAct and grounding
- Activity: search, then act

Lecture 3: Evaluating agents

- Ninety percent, then sixty
- pass@k and pass@1
- Intervals and noise
- Contamination and gaming
- Activity: audit an evaluation

Lecture 4: Safe agents

- One reminder, three thousand emails
- Reversibility and reach
- Least privilege and approval
- NIST AI RMF and the EU AI Act
- Activity: review the risks

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Problem set

1. Write PEAS for an exam-timetabling agent and classify its environment.

2. A two-state MDP: in s_1, action a gives reward 1 and stays; action b gives 0 and moves to s_2, where every step gives 2 forever. With $\gamma = 0.9$, which action is optimal in s_1?

3. Prove that straight-line distance is an admissible heuristic for road routing.

4. From n = 10 samples with c = 3 successes, compute the unbiased estimate of pass@1 and pass@2.

5. Two agents score 64% and 58% on 100 tasks. Is the difference significant at roughly 95%?

6. For a library-assistant agent with tools search, renew, pay_fine and email_patron, assign a control to each.

Solutions

> **1.** P: no clashes, room capacity met, spread for students. E: courses, rooms, enrolments. A: assign slots and rooms. S: enrolment and room databases. Fully observable, deterministic, sequential, static.

> **2.** V(stay) = 1/(1 - 0.9) = 10. V(move) = 0 + 0.9 × 2/(1 - 0.9) = 18. Action b is optimal.

> **3.** Any road path between two points is at least as long as the straight segment between them (triangle inequality), so h never exceeds the true cost.

> **4.** pass@1 = 1 - C(7,1)/C(10,1) = 0.3. pass@2 = 1 - C(7,2)/C(10,2) = 1 - 21/45 ≈ 0.533.

> **5.** SE of the difference ≈ √(0.64 × 0.36/100 + 0.58 × 0.42/100) ≈ 0.069; 1.96 × 0.069 ≈ 0.135 > 0.06, so not significant.

> **6.** search: allow. renew: allow for the requesting patron only. pay_fine: require approval and log. email_patron: limit to the requesting patron; bulk email denied.

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Lab handout: Lab 3: A tool-using agent

Goal

Build, evaluate and constrain a ReAct agent for room booking.

Steps

1. Implement two tools: rooms.free(day) and rooms.book(room, day).
2. Build a ReAct loop with a step limit and logged transcripts.
3. Run it five times on twenty held-out requests.
4. Report pass@1 with a 95% interval and classify failures.
5. Add an approval gate on booking and rerun.

Hand in

Code, transcripts, an evaluation table and a one-page risk review.

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Assessment rubric

Criterion	Excellent	Good	Developing	Not yet
Problem specification	Precise, with reward risks analysed	PEAS and MDP precise	PEAS stated	Vague
Implementation	Correct, logged and robust	Correct and logged	Runs	Does not run
Evaluation rigour	Plus error analysis and baseline	Held-out with intervals	Several runs	One run
Safety and accountability	Review, approval and owner	Risk review applied	Some limits	None

Reading list

Russell, S., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.
Agents, search and planning.

Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction (2nd ed.). MIT Press.
MDPs and RL.

Hart, P. E., Nilsson, N. J., & Raphael, B. (1968). A formal basis for the heuristic determination of minimum cost paths. IEEE Transactions on Systems Science and Cybernetics, 4(2), 100–107.
A*.

Yao, S., et al. (2023). ReAct: Synergizing reasoning and acting in language models. International Conference on Learning Representations (ICLR).
Tool-using agents.

Chen, M., et al. (2021). Evaluating large language models trained on code. arXiv:2107.03374.
pass@k and its estimator.

Amodei, D., et al. (2016). Concrete problems in AI safety. arXiv:1606.06565.
Reward hacking and safe exploration.

National Institute of Standards and Technology (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0). NIST AI 100-1.
Governance.

Saltzer, J. H., & Schroeder, M. D. (1975). The protection of information in computer systems. Proceedings of the IEEE, 63(9), 1278–1308.
Least privilege.